17.04.2024

EE2004 MICROPROCESSORS MIDTERM EXAM

Name:

CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.
DURATION: 60 min.

Q1) (20 points) a) What do the flags: N, Z, C and V stand for?

Negative Carry
Zero oVerflow

b) Consider the subtraction below,

```
LDR     R0, =0x20
LDR     R1, =0x10

SUBS    R2, R1,R0
```

b1) What will be the value of the N, Z, C and V bits?

N 1, Z 0, C 0, V 0

b2) What will be the result in Register R2 (in hex)?

0x FFFFFFF0.

b3) What will be the result in Register R2 (in signed decimal)?

-16

c1) What is result of 0x2F & 0x27?

0x27

c2) What is result of 0x2F | 0x27?

0x2F

c3) What is result of 0x2F ^ 0x27?

0x8

c4) What is result of 0x2F >> 3?

0x5

c5) What is result of 0x27 << 4?

0x270

Q2) (15 points) Write an assembly description that performs the given C code.

```
if(Data >= 32){
  Data = 255; // ceiling
}else{
  Data = Data<<3;
}
```

```
MOV   R0, #data
CMP   R0, #32
MOUGE R0, #255
LSLLT R0, R0, #3
```

Q3) (10 points) What are the power modes supported by Stm32f4 controllers? Briefly explain.

ACTIVE (RUN) MODE : No power saving property. All peripherals and clock sources can be activated.

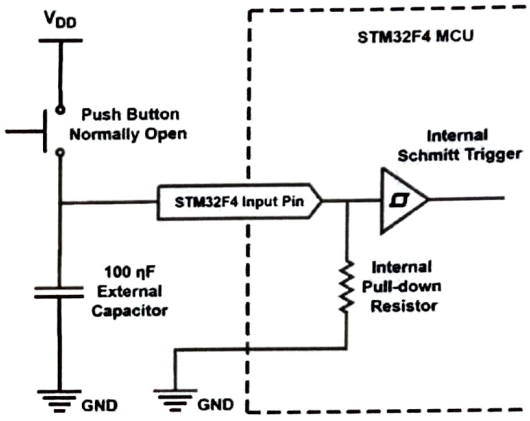
SLEEP MODE: CPU is stopped in this mode. SRAM and peripherals are active.

STOP MODE: CPU can be woken up by external interrupts. SRAM and peripherals (registers) retain the previous contents.

STANDBY MODE: CPU can exit from this mode by resetting. Only RTC and small portions of SRAM and registers can be active.

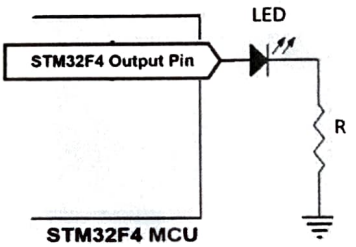
Q4) (10 points) Explain purpose and operation of the following circuit.

This is the hardware solution to avoid switch bouncing. The circuit is composed of a low-pass RC filter and a Schmitt trigger. Internal resistance values are typically 40k. Together with the external capacitor RC filter is formed that filters out switch bouncing.



Q5) (10 points) Interface an LED to a selected output pin as given in the figure. The desired LED operating point is 2.4V, 1mA. The microcontroller output high voltage is 3.3V. Find required resistor value R.

$$\frac{3.3 - 2.4}{1m} = 0.9\,k\Omega$$



Q6) (10 points) A standalone embedded system is to be designed. As power source, a 1800 mAh battery is selected. Embedded system spends 10 seconds in run mode in every ten minutes and remaining time in sleep mode. Run and sleep modes draw 150 mA and 10 mA respectively. Calculate the time of battery change.

$\frac{1500}{600} + \frac{5900}{600} = \frac{7400}{600} = 12.33\,mA$ $\quad 1800/12.33 = 145.985 \sim 146$ hours

$\frac{150}{60} + \frac{600}{60} = \frac{750}{60} = 12.5\,mA$ $\quad 1800/12.5 = 144$ hours

Q7) (15 points) Fill addresses and values after the assembly code at the left.

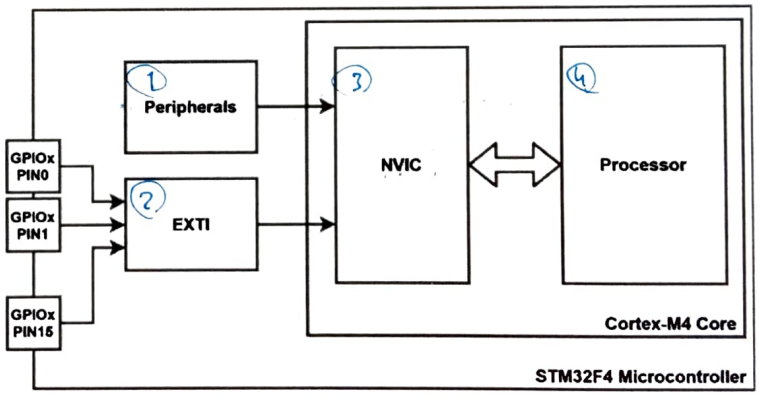
```
Loc1    EQU     0x20000400

        LDR     R0, =Loc1
        LDR     R1, =0xF0F1F2F3
        MOV     R2, #4
        LDR     R3, =0x04040404

loop1   STR     R1, [R0]
        SUB     R2, R2, #1
        ADD     R0, R0, #4
        ADD     R1, R1, R3
        CMP     R2, #0
        BGT     loop1
```

Address	Value
0x20000400	0xF0F1F2F3
0x20000404	0xF4F5F6F7
0x20000408	0xF8F9FAFB
0x2000040C	0xFCFDFEFF
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Q8) (10 points) Explain modules that handle interrupts in Stm32f4 processors utilizing the diagram below.



(1) First block contains peripheral units that can generate interrupt requests.

(2) Within the second block, GPIO pin interrupts are handled by the external interrupts and events controller.

(3) In the third block all interrupt requests are handled by the NVIC.

(4) Accepted interrupt requests are processed by the CPU.